Primary 3 · Science

# Science · Set 5

80 marks · 1 h 15 min · Challenge

Name: Class: Score: / 80

## Section A: Multiple-Choice Questions (40 marks)

*Choose the correct answer. Each question is worth 2 marks.*

**1.** A pupil pours the same amount of water into three identical pots, each filled with a different type of soil — sandy soil, clay soil, and loamy soil — and times how long the water takes to fully drain through each pot. What was CHANGED in this test? ( )

(1) The type of soil (2) The amount of water poured (3) The size of the pot
(4) The person pouring the water

**2.** What was KEPT THE SAME in this test? ( )

(1) The amount of water poured and the size of the pot (2) The type of soil
(3) How long the water took to drain (4) The colour of the soil

**3.** The table shows how long the water took to drain through each type of soil. Based on the table, which soil drained the water FASTEST? ( )

| Soil | Time to drain (s) |
|---|---|
| Soil A | 20 |
| Soil B | 95 |
| Soil C | 55 |

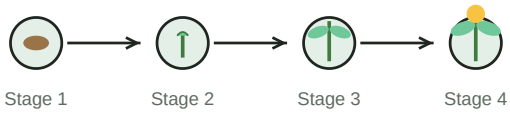
(1) Soil A (2) Soil B (3) Soil C (4) All drained at the same speed

**4.** What is the one thing in this table that was changed between Soil A, Soil B and Soil C? ( )

| Soil | Time to drain (s) |
|---|---|
| Soil A | 20 |
| Soil B | 95 |
| Soil C | 55 |

(1) The amount of water used (2) The size of the pot (3) The time of day
(4) The type of soil

**5.** Look at the stages of a bean plant's life cycle. Which of these happens right after the adult plant produces flowers and fruits, starting the cycle again? ( )




(1) Seedling (2) Young plant (3) New seeds (4) Roots

**6.** Two identical seeds are both given water and air, but Seed X is kept at a warm room temperature while Seed Y is kept inside a freezer. Which seed is most likely to germinate? ( )

(1) Seed X (2) Seed Y (3) Both will germinate equally well
(4) Neither will germinate

**7.** Which of these is a living thing? ( )

(1) A dried flower in a vase (2) A stuffed toy animal (3) A photograph of a dog
(4) A pot of basil growing on a windowsill

**8.** A bat is covered in fur and feeds its young with milk. Which group of animals does it belong to? ( )

(1) Birds (2) Insects (3) Mammals (4) Reptiles

**9.** Which of these animals does NOT hatch from an egg? ( )

(1) Chicken (2) Butterfly (3) Guinea pig (4) Lizard

**10.** A pupil notices that a damp towel left bunched up in a gym bag for a week starts to smell and develops mould. Which condition most likely helped the mould to grow? ( )

(1) The towel was kept completely dry (2) The towel was left in direct sunlight
(3) The towel was frozen (4) The towel was kept damp and in the dark

**11.** Which TWO animals in this list both have a life cycle with a pupa stage? ( )

(1) Cockroach and grasshopper (2) Chicken and frog (3) Mosquito and beetle
(4) Cockroach and chicken

**12.** A dragonfly's life cycle (egg, nymph, adult) is most similar to which of these animals? ( )

(1) Butterfly (2) Mosquito (3) Cockroach (4) Beetle

**13.** Which of these is the correct order of a frog's life cycle, starting from the egg? ( )

(1) Egg → Froglet → Tadpole → Adult frog (2) Egg → Adult frog → Tadpole
(3) Tadpole → Egg → Adult frog (4) Egg → Tadpole → Tadpole with legs → Adult frog

**14.** Which stage of a beetle's life cycle looks most different from the adult beetle? ( )

(1) Egg (2) Larva (3) Pupa (4) Adult

**15.** A canvas awning at a market stall needs to keep rain off the stalls below. Which property is MOST important for the awning's material? ( )

(1) Transparent (2) Waterproof (3) Magnetic (4) Edible

**16.** Which of these would be the best choice for making a flask to keep a drink cold or hot for several hours? ( )

(1) Thin paper cup (2) Double-walled insulated steel (3) Clear plastic cup
(4) Open ceramic bowl

**17.** Which object would float on water due to trapped air inside it, even though it is made of a material that can sink? ( )

(1) A solid metal cube (2) A wet paper towel (3) A smooth pebble
(4) A closed empty plastic bottle

**18.** Which of these is a non-living thing? ( )

(1) A growing crystal (2) A growing seedling (3) A growing puppy
(4) A growing colony of bacteria

**19.** Which of these is TRUE about both birds and fish? ( )

(1) Both are covered in scales (2) Both hatch from eggs (3) Both can fly
(4) Both live only in water

**20.** Bread dough rises when yeast is added because the yeast... ( )

(1) Absorbs water and swells up like a sponge
(2) Is a living fungus that feeds and produces gas as it grows
(3) Reacts chemically like baking soda (4) Is heated and expands like a balloon

## Section B: Structured Questions (16 marks)

*Write a short answer, or tick True or False, for each question. Each question is worth 2 marks.*

21. Tick True or False for each statement. Write True or False.

| Sentence | True / False |
|---|---|
| (a) Spraying water on a wilting plant to perk it up is a way to TEST whether it is alive. | |
| (b) Checking whether a plant's stem slowly bends towards a light source over a few days is a way to TEST whether it is alive. | |

**22.** A pupil pushes the same toy car on the same flat floor three times, using a gentle push, a medium push, and a hard push, and measures how far it rolls each time. What was CHANGED in this test? ( )

(1) The force used to push the car (2) The toy car used (3) The floor the car rolled on (4) The distance the car rolled

**23.** Give one property that makes thick cloth suitable for making an oven mitt. (2 marks)

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**24.** Give one property that makes silicone suitable for making an ice cube tray that is easy to pop the ice cubes out of. (2 marks)

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**25.** A pupil notices mould growing on an orange left in a fruit bowl for two weeks. Which condition most likely helped the mould grow? ( )

(1) The orange was kept in a freezer the whole time

(2) The orange's skin slowly broke down and stayed moist

(3) The orange was completely sealed in an airtight dry jar

(4) The orange was kept under strong, direct sunlight

26. Tick True or False for each statement. Write True or False.

| Sentence | True / False |
|---|---|
| (a) All insects go through a pupa stage in their life cycle. | |
| (b) A mosquito's larva and pupa stages both live in water. | |

27. Tick True or False for each statement. Write True or False.

| Sentence | True / False |
|---|---|
| (a) An object that traps air inside it is more likely to float. | |
| (b) A material cannot be both strong and flexible at the same time. | |

**28.** Name ONE characteristic of living things that a growing crystal does NOT have. (2 marks)

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## Section C: Open-Ended Questions (24 marks)

*Explain your answer in full sentences. Each question is worth 3 marks.*

**29.** A pupil wants to find out which material — a sponge, a kitchen cloth, or a plastic sheet — absorbs the most water. She cuts each material to the same size, dips each one into the same amount of water for the same length of time, then weighs how much water each one has absorbed. Describe what was changed, what was kept the same, and why this makes it a fair test. (3 marks)

**30.** A pupil wraps identical ice cubes in newspaper, foam, and cling film, and leaves them at the same room temperature. She records how long each ice cube takes to melt completely. Describe what was changed, what was kept the same, and why this makes it a fair test. (3 marks)

**31.** A pupil finds a snail stuck to the inside of a flower pot, not moving. Describe TWO tests she could do to find out whether it is alive, other than just looking at it. (3 marks)

**32.** A pupil leaves a damp sponge in a closed, dark container for a week. She finds mould growing on it. Explain where the mould's food comes from, and why keeping food covered and dry at home helps to stop mould from growing. (3 marks)

**33.** A dragonfly's life cycle has 3 stages: egg, nymph, adult. A beetle's life cycle has 4 stages: egg, larva, pupa, adult. Explain ONE difference between these two life cycles. (3 marks)

**34.** A pupil wants to choose a material to make a flask that keeps a drink hot for many hours. Explain why a double-walled steel flask would be a good choice, giving TWO properties that make it suitable. (3 marks)

**35.** Farid says that the mould growing on his bread and the green algae growing in his fish tank are the same kind of living thing because both appear without being planted. Explain the KEY difference between fungi (like mould) and plants (like algae). (3 marks)

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**36.** A pupil plants two identical seeds, giving both the same amount of water and air. She keeps Pot X at room temperature and places Pot Y inside a refrigerator. After a week, the seed in Pot X has germinated, but the seed in Pot Y has not. Explain why this happened. (3 marks)

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**End of paper. Check your answers.**